# 18 CAPACITANCE

18.1 Capacitors and Capacitance

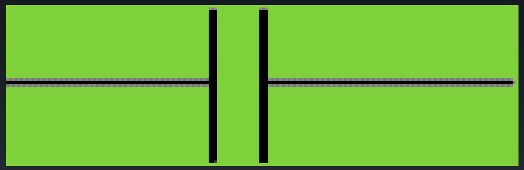
Function

Define

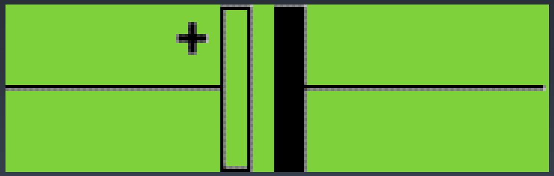
Capacitors in series and parallel

18.2 Energy stored in a capacitor

# Capacitor symbols



unpolarised capacitor symbol



polarised capacitor symbol

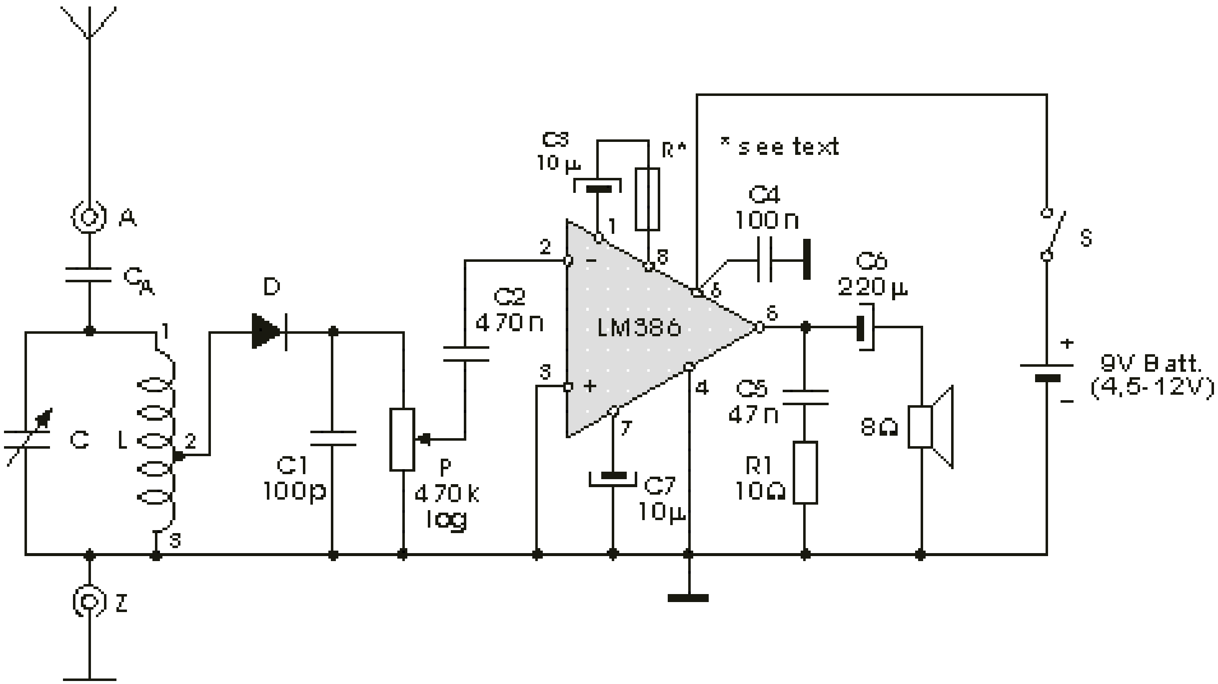
variable capacitor

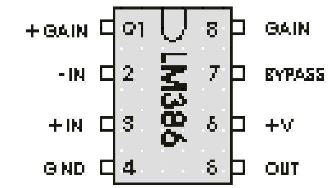
http://www.kpsec.freeuk.com/capacit.htm

# Uses

- Components in electronics and telecommunication.
- Radio and television receivers and in transmitter circuits.
- In power supplies to smooth the rectification of a.c. into d.c.
- In circuits where currents are switched.

# Radio receiver



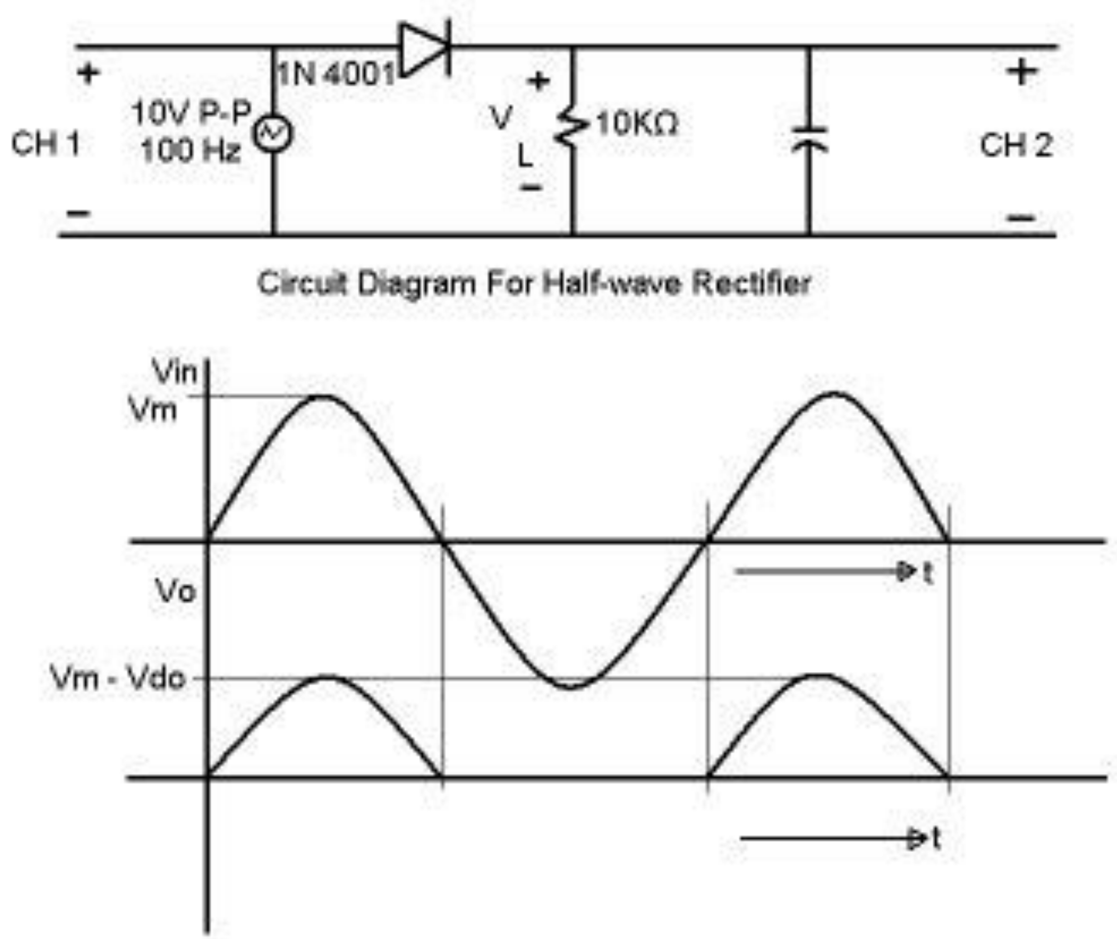




+GAIN, GAIN : By connecting a 10mF capacitor (plus pole on +GAIN) and resistor between these two pins, voltage amplification can be set to any value from 20 to 200.
+IN - non-inverting input. Input and output voltages are in phase.
-IN - Inverting input. Input and output voltages are in counter-phase.
BYPASS - override
+V- positive power supply input
GND - ground, negative power supply input
OUT - output

*Pic. 3.18 a- Electronic diagram of the detector (TRF) radio receiver with audio amplifier built around the LM386 IC*

http://www.mikroe.com/old/books/rrbook/chapter3/chapter3d.htm

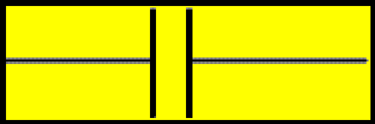
In power supplies to smooth the rectification of a.c. into d.c.



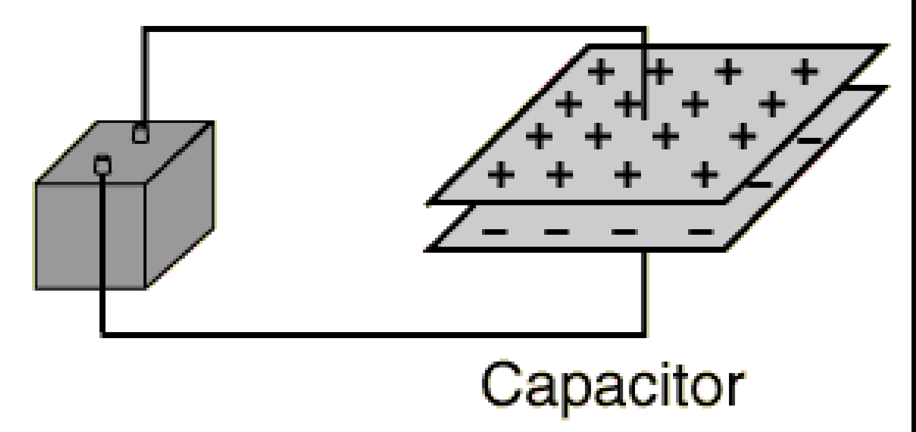

Half Wave Rectifier Circuit and Wave Front

http://electronics-lab-experiment.blogspot.com/2009/02/experiment-study-of-diode-rectifier.html

# Capacitor



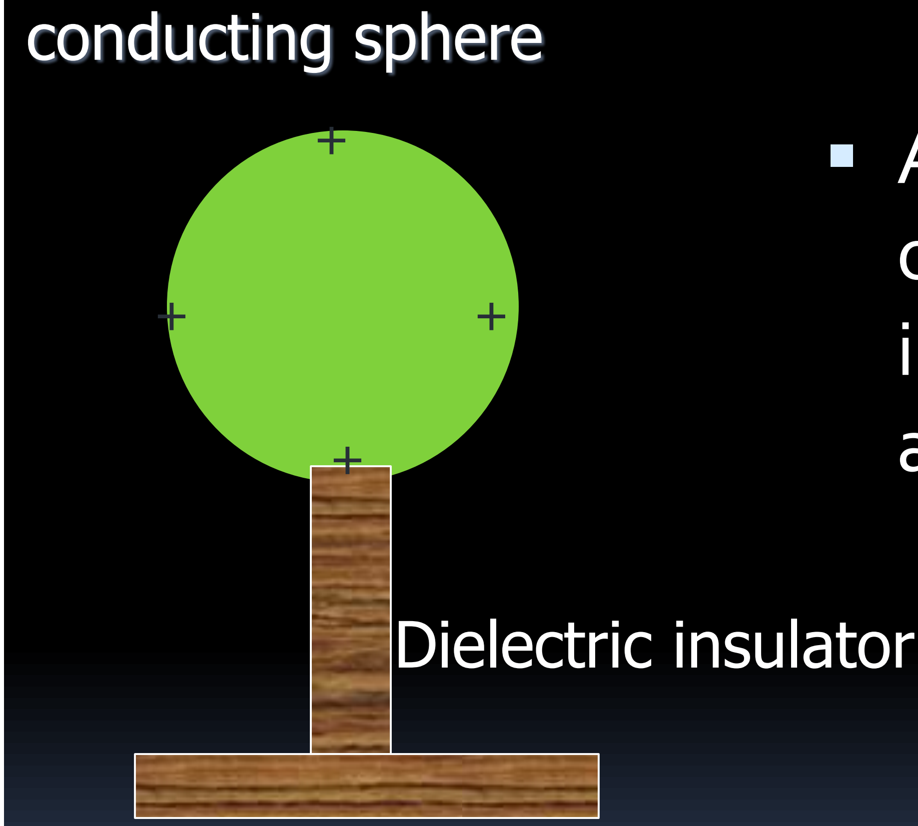
symbol




- **A capacitor** or condenser is a device for storing charge e.g. parallel conducting plates
- Capacitance is typified by a parallel plate arrangement with a dielectric in between the plates.
- A **dielectric** is an insulating material, it can be air, oil, mica, polystyrene or paper. Dielectric between the parallel plates increases the capacitance of the capacitor.

hyperphysics.phy-astr.gsu.edu/.../potsph.html

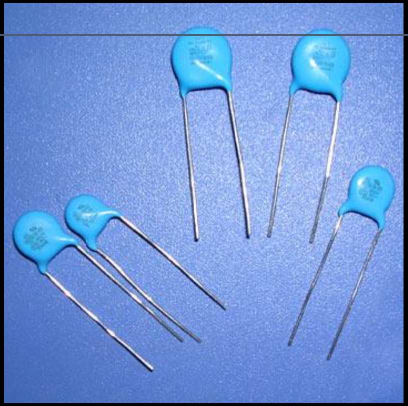
# Capacitor: Charged conducting sphere



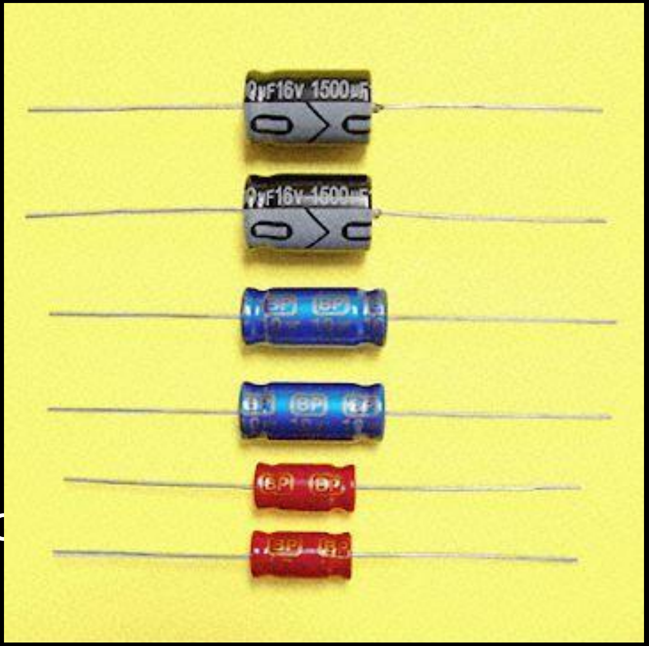

- Any capacitor is make up of two conductors insulated from one another.

# Practical Capacitors

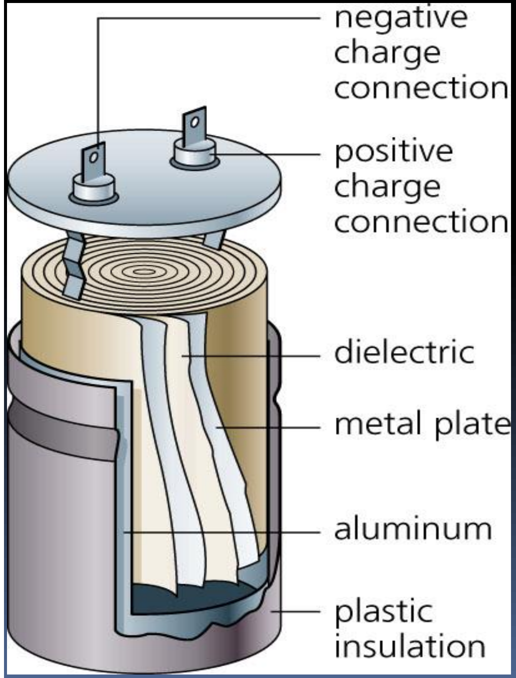
Tantalum Capacitor

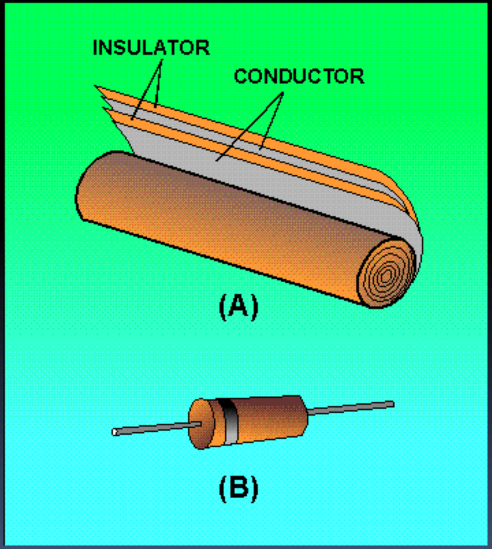


High Voltage Ceramic Capacito



Aixal Type Electrolytic **Capacitor**



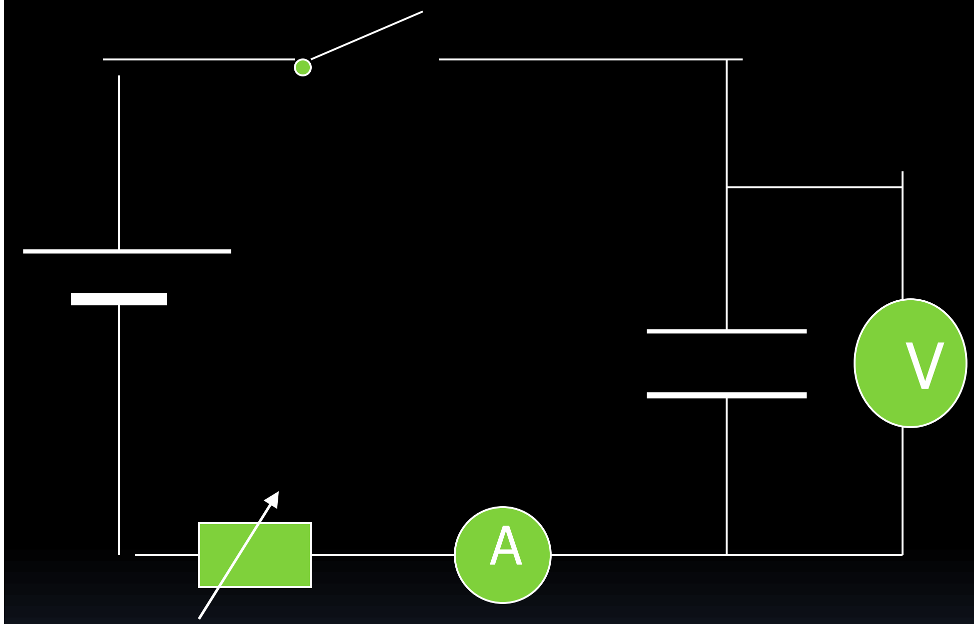

Paper capacitor

# *Charging a capacitor.*

When a capacitor is connected to a cell, the cell provides a p.d. across the plates, electrons from the negative terminal of the cell flows to the negative plate, and repels the electrons from the positive plates, the electrons move to the positive terminals of the cell leaving the positive plate positively charged. As opposite charges accumulates on the plates the potential difference across the plates increases. Charges stop flowing when the p.d. across the plates equals the p.d. of the cell.

*Experiment.*

Aim: To investigate the charge stored in a capacitor.




Procedure:

a) When the switch is closed, electrons pass along the wires to charge the capacitor.

b) The microammeter indicates the current due to the flow of charge.

c) The p.d. across the capacitor is measured using a high resistance voltmeter, connected across it.

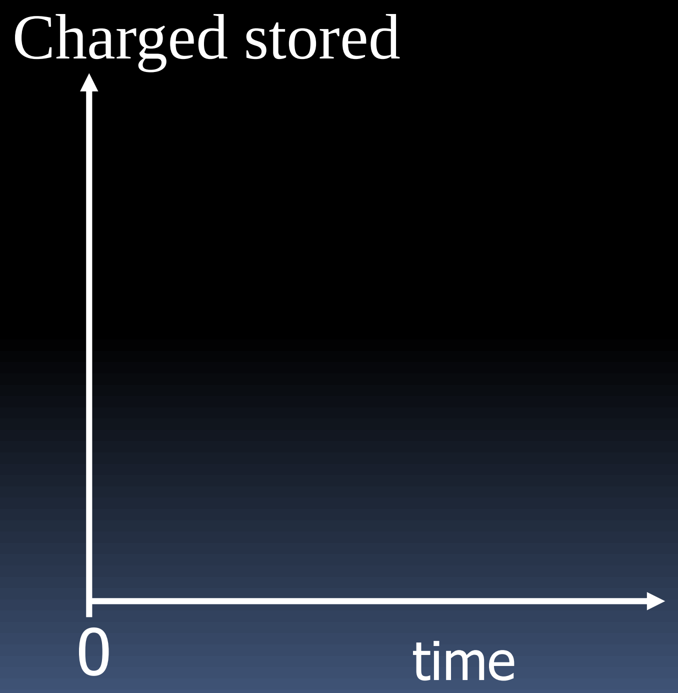
## *Experiment.*

d) The variable resistor is initially set at high resistance. It is adjusted to keep the current constant, until the capacitance is fully charged.

e) When the switch is closed, a stop watch is started and the voltmeter reading is noted at regular intervals.

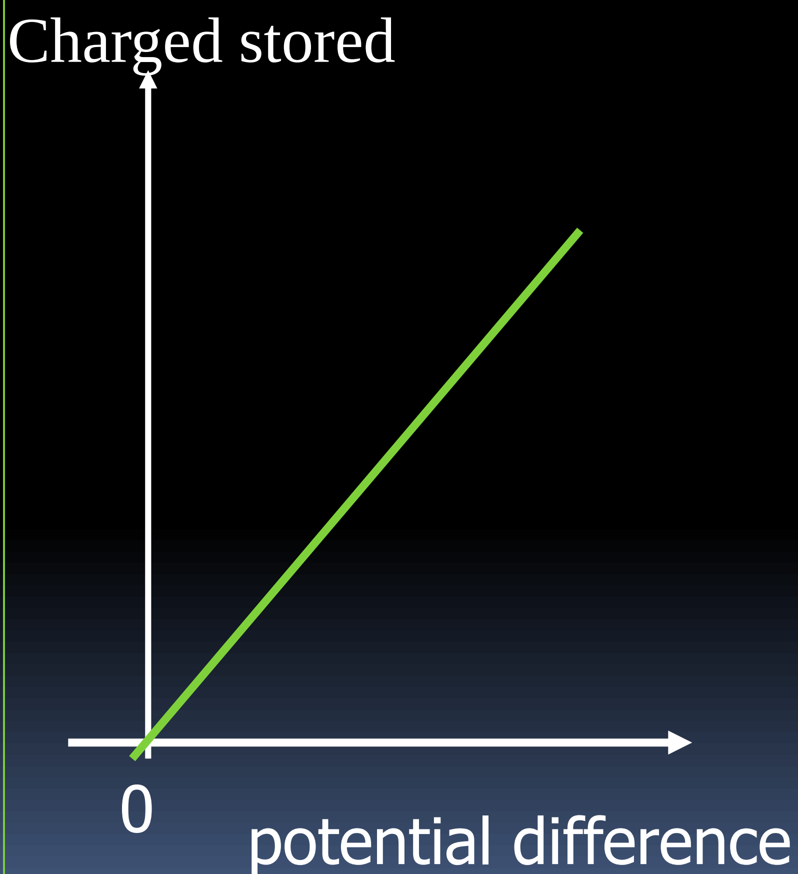
f) The charged stored, Q in the capacitor at any time t is, $Q = It$

| | | | | | |
|---|---|---|---|---|---|
| t/s | | | | | |
| Q/C | | | | | |
| V/V | | | | | |

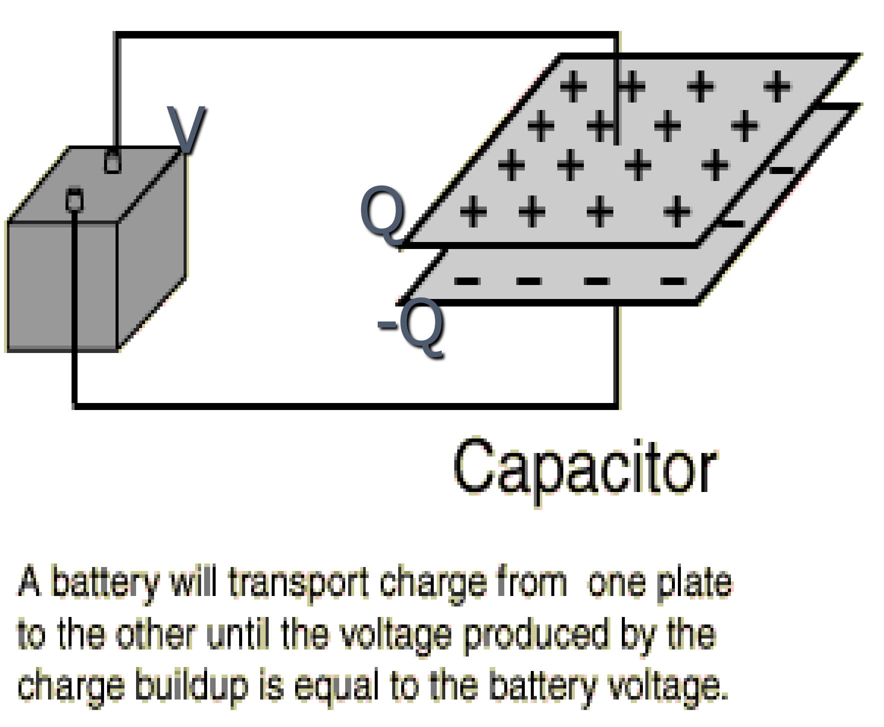


# Charges stored and p.d. across capacitor when charge with constant current.

- Plot a graph of Q against V.
- The graph of Q against V is a straight line
- $Q = mV$
- i.e. is p.d. is directly proportional to the charge stored.
- As charged stored on the capacitor increase the p.d. across it increases.



# Definition




The **capacitance** of a capacitor is the amount of charged stored on the plates per unit potential difference between the plates.

$$C = \frac{Q}{V}$$

where

Q = magnitude of charge stored on each plate.

V = voltage applied to the plates.

Unit: $C\,V^{-1}$ = farad (F)

hyperphysics.phy-astr.gsu.edu/.../potsph.html

# Measurement of capacitance


The second method to test capacitor is to use digital capacitance meter and is a little more accurate compares to analog multimeter. Connect the test probe to the capacitor and read the result from the meter lcd display. Example, a 100 microfarad should have the reading of somewhere 90 microfarad to 110 microfarad. Remember, capacitors have tolerance just like resistors.

www.electronicrepairguide.com/test-capacitor.html

# Capacitance

a) Or **Q = C.V**

b) One farad is the capacitance of a capacitor when the charged stored is 1 C when the p.d. across it is 1 volt.

$1\ \mu F = 1 \times 10^{-6}\ F$

$1\ pF = 1 \times 10^{-12}\ F$

Briefly explain how a capacitor stores energy.

- Basically a capacitor stores opposite charges on the two parallel plates a distance apart. Thus we can imagine that each electron is separate from its positive ions, in doing the separation work is required against the attractive forces. This work is stored as electrical potential energy in the electric field.

# Example 18.1

If the radius of the spherical conductor is r, what is the capacitance of the conductor?

Solution

If the spherical conductor has a charge Q, then the potential on the surface is

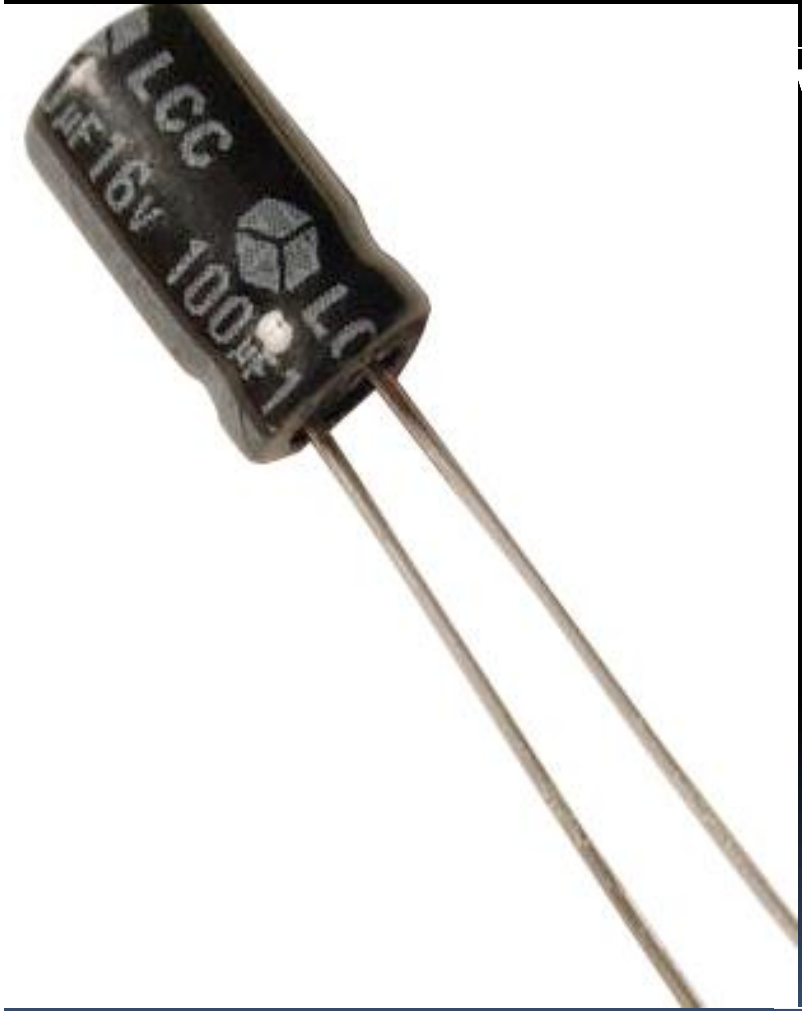
$$V = \frac{Q}{4\pi\varepsilon_o r}$$

Since

$$C = \frac{Q}{V}$$ (definition)

$= 4\pi\varepsilon_0 r$

# Working voltage

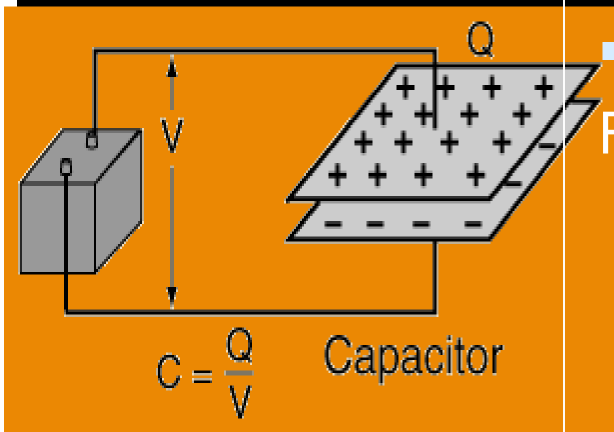



**Working voltage** is the maximum voltage that can be applied across the capacitor safely. For voltages above this value the capacitor breaks down. This value is written on the capacitor beside the value of the capacitance of the capacitor.

# Energy Stored on a Capacitor

- The energy stored on a capacitor can be calculated from the equivalent expressions:
- This energy is stored in the electric field.

From the definition of voltage as the energy per unit charge, one might expect that the energy stored on this ideal capacitor would be just QV. That is, all the work done on the charge in moving it from one plate to the other would appear as energy stored. But in fact, the expression above shows that just half of that work appears as energy stored in the capacitor. For a finite resistance, one can show that half of the energy supplied by the battery for the charging of the capacitor is dissipated as heat in the resistor, regardless of the size of the resistor.




$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV = \frac{1}{2}CV^2$$

# Energy Stored on a Capacitor (derivation: for info)

When a small positive charge δQ is transferred from the negative plate to the positive plate (the charge is so small that it does not affect the potential between the plates), work is done against the attractive force on the negative plates.

$\delta W = V.\,\delta Q$

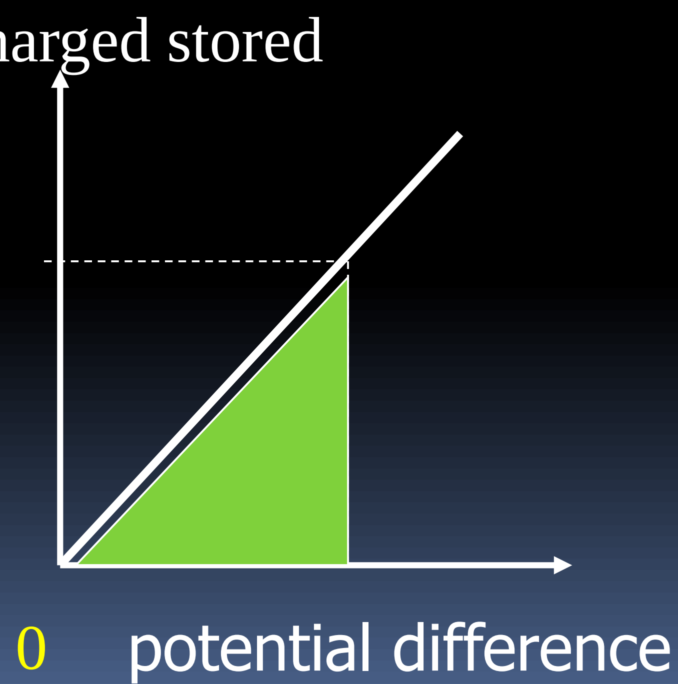
If the capacitor is charged from zero to a final charge of Q, then

$$W = \int_0^Q V.dQ$$

as V=Q/C

= area under graph

= $^1/_2$ Q.V



# Example 18.2

An uncharged 0.10 μF capacitor is charged to a p.d. of 500 V by a battery. Calculate

a) the energy stored in the capacitor,

b) the charged circulated by the battery,

c) the energy provided by the battery,

d) the total heat dissipated in the resistance of the connecting wires and of the battery.

(Ans. A) $1.25 \times 10^{-2}$ J; b) $5.0 \times 10^{-5}$ C; c) $2.5 \times 10^{-2}$ J; d) $1.25 \times 10^{-2}$ J)

a) $W = \frac{1}{2} CV^2$

$= \frac{1}{2} (0.1 \times 10^{-6}) 500^2$

$= 1.25 \times 10^{-2}$ J

b) $Q = CV$

$= (0.1 \times 10^{-6}) 500$

$= 5.0 \times 10^{-5}$ C

c) $W_s = QV$

$= 5.0 \times 10^{-5} (500)$

$= 2.5 \times 10^{-2}$ J

d) heat dissipated

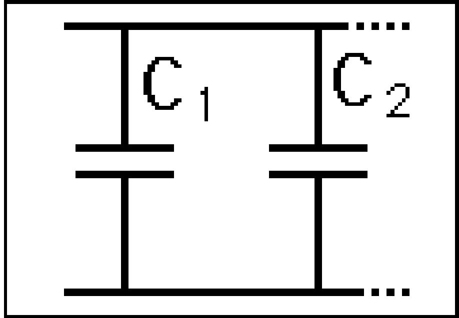
$= W_s - W$

$= 2.5 \times 10^{-2} - 1.25 \times 10^{-2}$

$= 1.25 \times 10^{-2}$ J

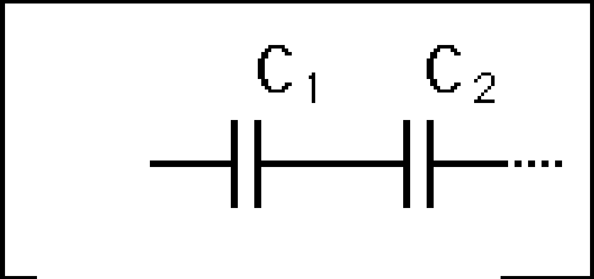
# Capacitor Combinations

Capacitors in parallel add ...




$$C_{eq} = C_1 + C_2 + \ldots$$

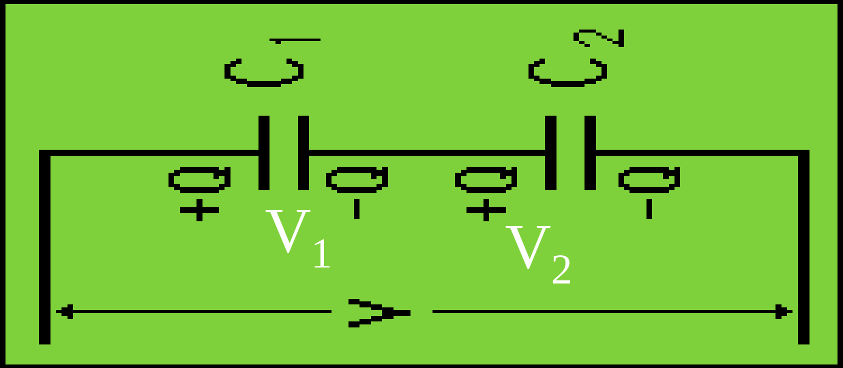
Capacitors in series combine as reciprocals ...

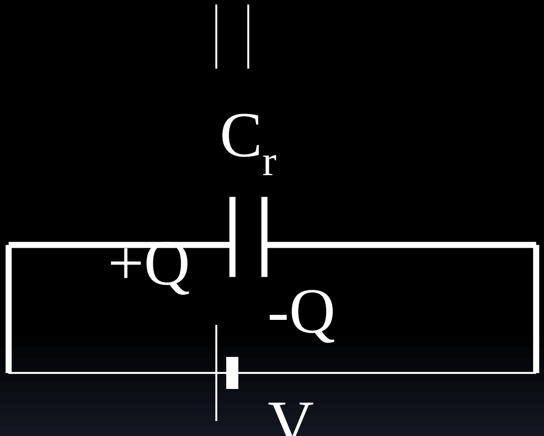



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$$

- Note opposite to resistance.

# Capacitors in series




a) The charge on the right hand plate is the same. The equivalent capacitor stores the same charge Q and the p.d. across it is the same as the p.d. across the series capacitors.

b) The reciprocal of the resultant (combined, effective) capacitance $C_r$ is equal to the sum of the reciprocal of the individual capacitance.

$1/C_r = 1/C_1 + 1/C_2 + 1/C_3$

$V = V_1 + V_2 + V_3$ as $V = Q/C$

$$\frac{Q}{C_r} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

# Capacitors in series

a) The charge on the right hand plate is the same. The equivalent capacitor stores the same charge Q and the p.d. across it is the same as the p.d. across the series capacitors.

b) The reciprocal of the resultant (combined, effective) capacitance $C_r$ is equal to the sum of the reciprocal of the individual capacitance.

$$1/C_r = 1/C_1 + 1/C_2 + 1/C_3$$

a) The effective capacitance is always smaller than the smaller capacitance.

b) The effective capacitance increases when a capacitance in the series network is removed.

c) For n identical capacitors in series, each of capacitance C, the effective capacitance is

$$C_r = C/n$$

# Charge on Series Capacitors

Since charge cannot be added or taken away from the conductor between series capacitors, the net charge there remains zero. As can be seen from the diagram, that constrains the charge on the two capacitors to be the same in a DC situation. This charge Q is the charge you get by calculating the equivalent capacitance of the series combination and multiplying it by the applied voltage V.

- You store less charge on series capacitors than you would on either one of them alone with the same voltage!
- Does it ever make sense to put capacitors in series? You get less capacitance and less charge storage than with either alone. It is sometimes done in electronics practice because capacitors have maximum working voltages, and with two "600 volt maximum" capacitors in series, you can increase the working voltage to 1200 volts.

http://hyperphysics.phy-astr.gsu.edu/hbase/electric/capac.html

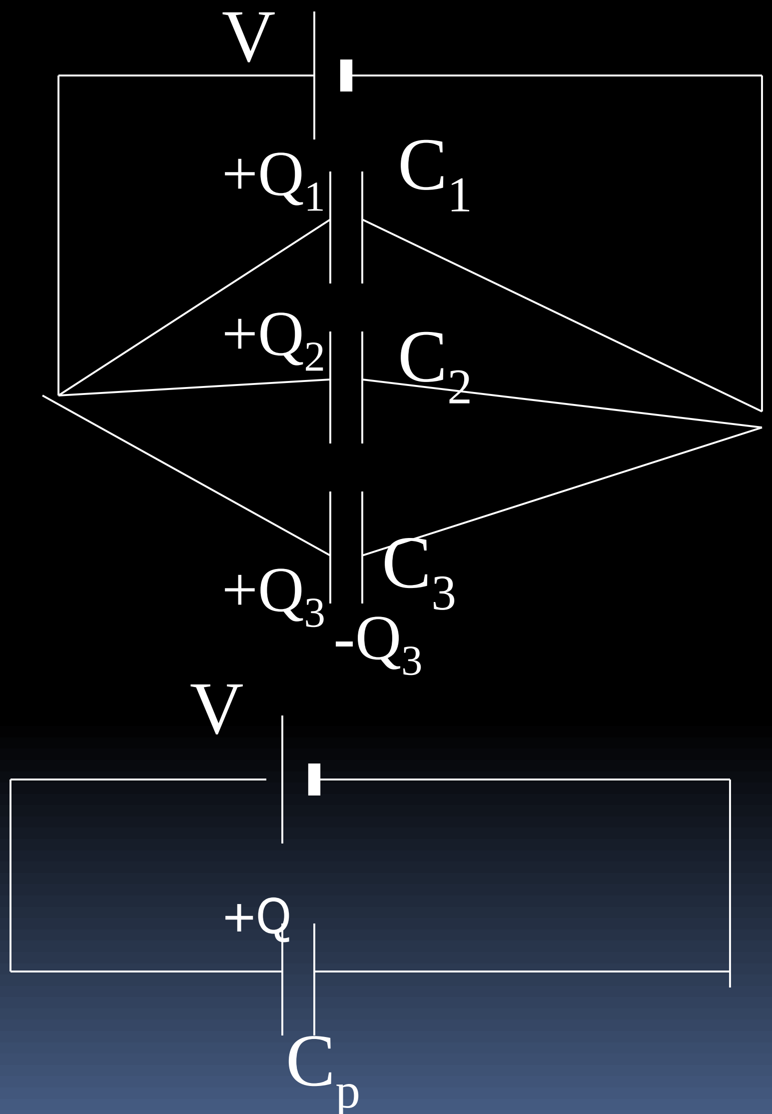
# Capacitors in parallel

The three capacitor can be replaced by an equivalent capacitor join to the same battery and stored the total charges stored in the capacitors.

The charges on all left-hand plate are positive.

$Q = Q_1 + Q_2 + Q_3$

a) The p.d. across each capacitors are the same.



# Capacitors in parallel

Since $Q = CV$

Then,

$C_p V = C_1 V + C_2 V + C_3 V$

Dividing through by V,

$C_p = C_1 + C_2 + C_3$

b) The effective capacitance $C_p$ is the sum of the individual capacitance.

$C_p = C_1 + C_2 + C_3$

c) For n identical capacitors in series each of capacitance C the effective capacitance is

$C_p = nC$

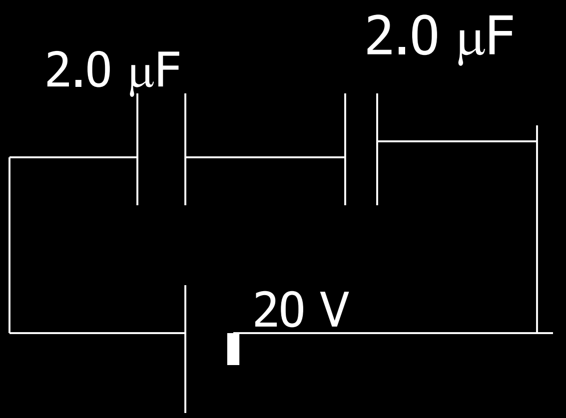
# Example 17.3

Given a number of capacitors each with a capacitance of 2.0 μF and a maximum safe working potential difference of 10V, how would you construct capacitors of

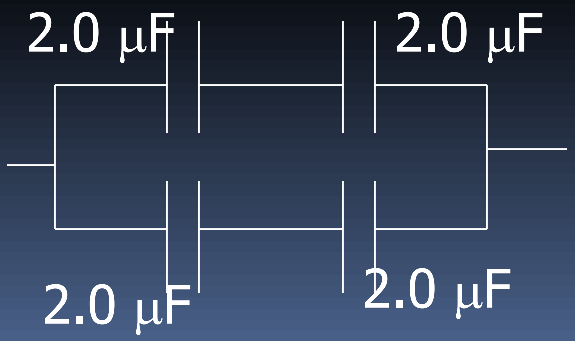
a) 1.0 μF capacitance, suitable for use up to 20 V,

b) 2.0 μF capacitance, suitable for use up to 20 V.

a) In series you take the reciprocal to obtain the resultant




b) Parallel and series

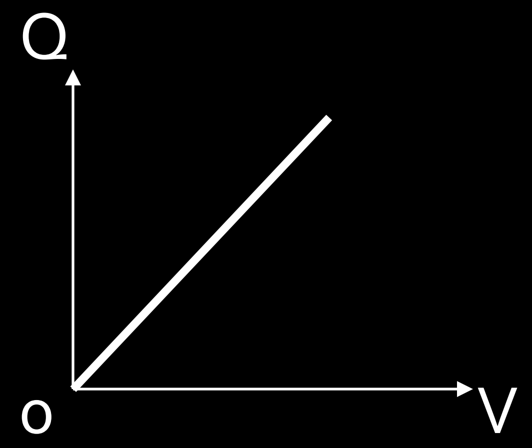


# Self test 18

1) Define capacitance
2) Sketch the graph to show how charge on capacitor varies as p.d. across it increases.

a) What does the gradient represents?
b) What does the area under graph represents?

1) Ratio of charge on either plates to the p.d. across the plates.
2) 




a) From C= Q/V

then, Q = C V, thus gradient is the capacitance

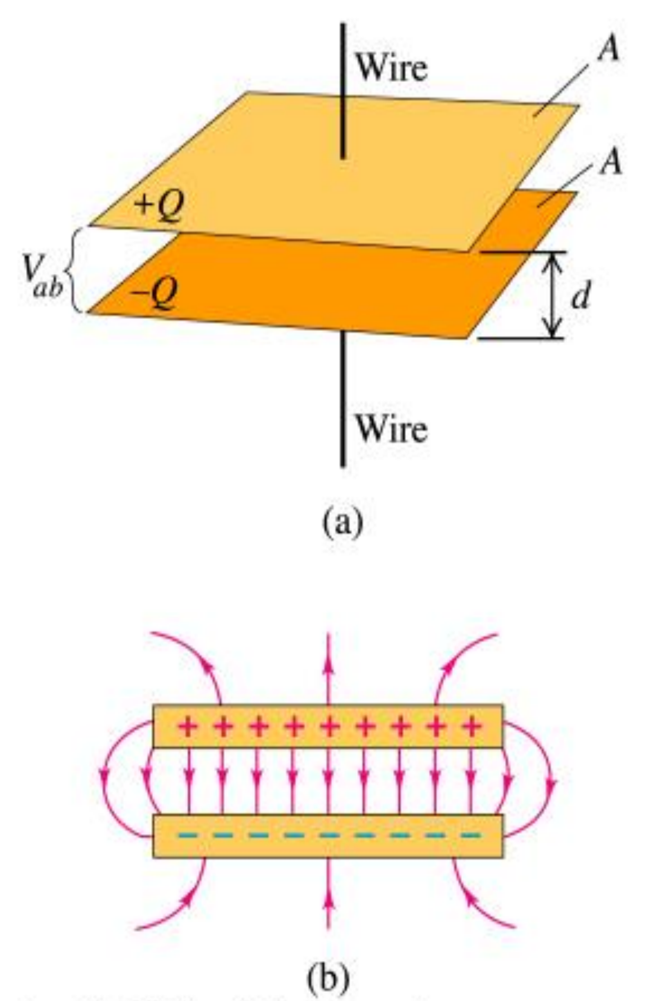
b) area under graph is the energy stored in capacitor

## Self test 18

3) If a capacitor of capacitance C is connected to a battery and the p.d. across the capacitor is V, what is a) the energy supplied and b) the energy stored in capacitor.

3) a) $CV^2$ b) $\frac{1}{2} CV^2$

# Factors affecting capacitance [info]





The capacitance of a parallel plates capacitor depends on

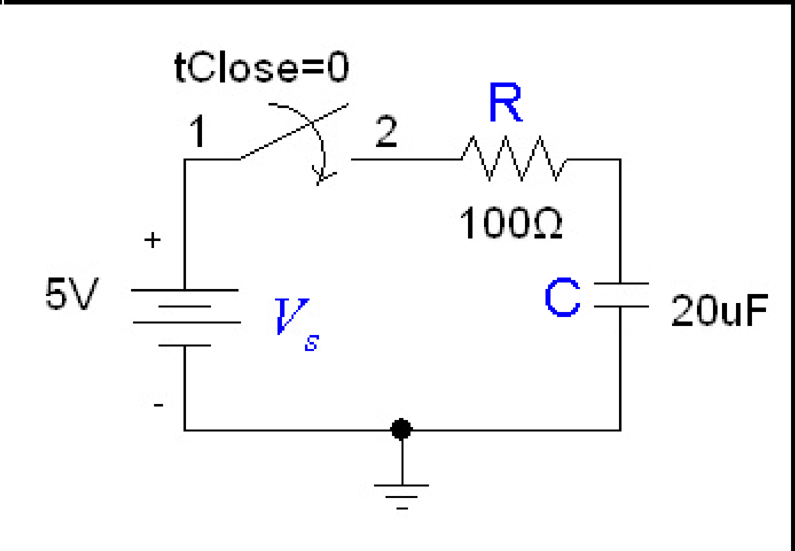
$$C = \frac{\varepsilon A}{d}$$

where A = cross-sectional area of each plates

d = separation

ε = permittivity of the material

# Charging




- the resistor in the circuit slows down the charging of the capacitor.
- initial current

$I_o = V/R = 5/100$

$= 50$ mA

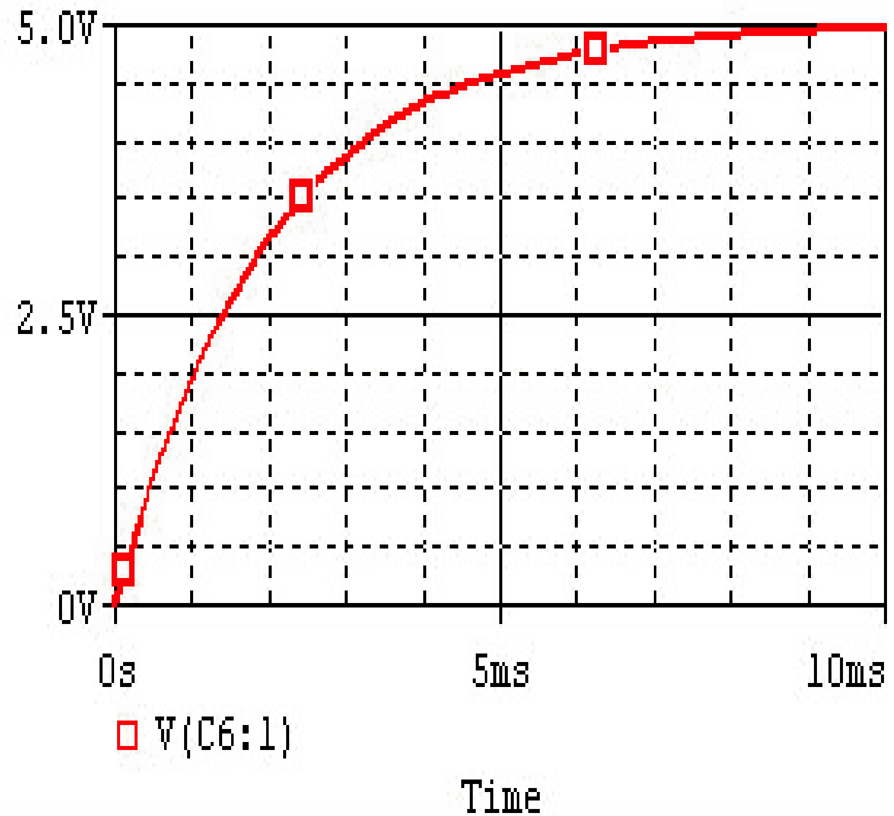
If the capacitor is initially uncharged and we want to charge it with a voltage source $V_s$ in the RC circuit:

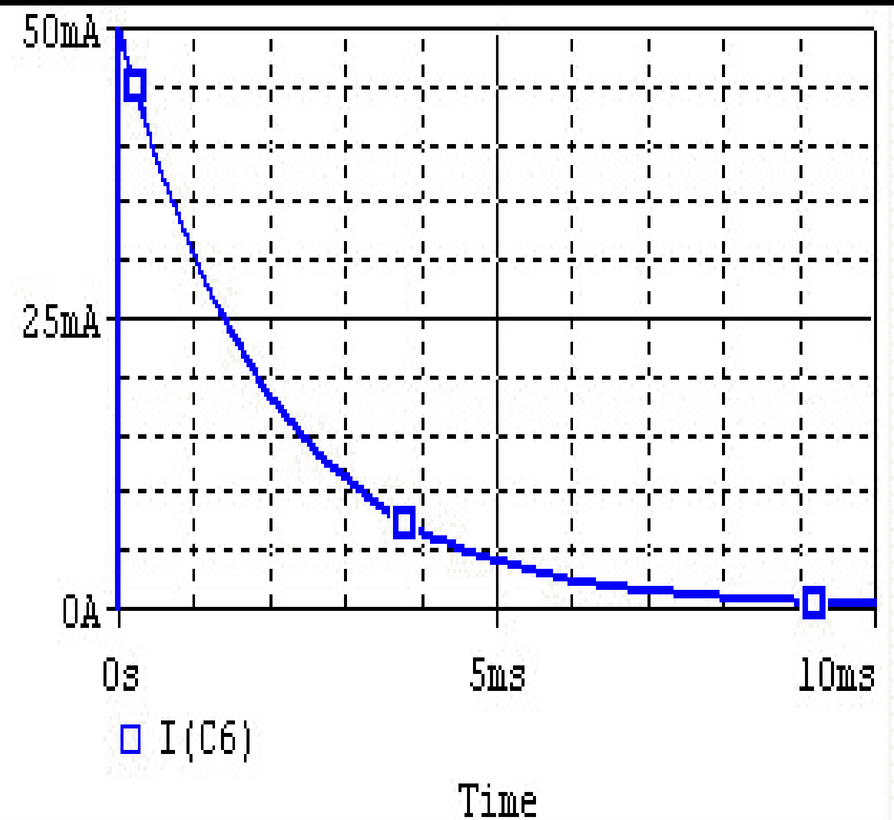
Current flows into the capacitor and accumulates a charge there. As the charge increases, the voltage rises, and eventually the voltage of the capacitor equals the voltage of the source, and current stops flowing. The voltage across the capacitor is given by:

$$v_C(t) = V_0(1 - e^{-\frac{t}{RC}})$$

where $V_o = V_S$, the final voltage across the capacitor

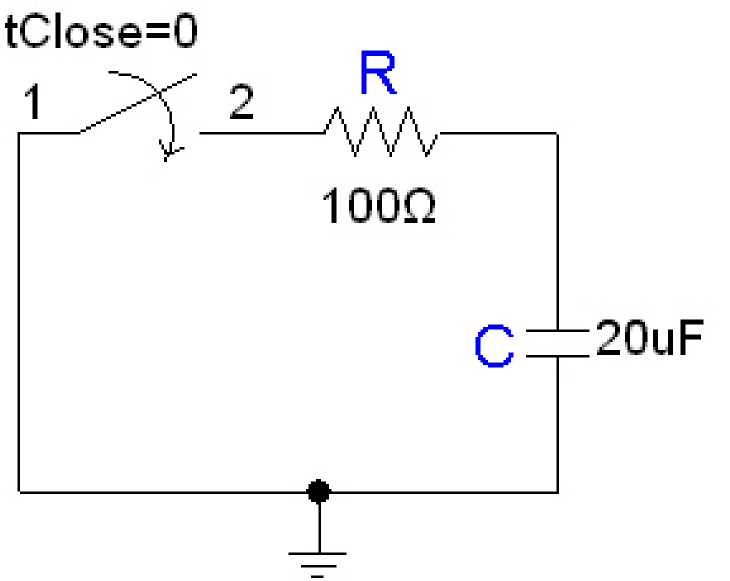
# Charging




http://hades.mech.northwestern.edu/index.php/RC_and_RL_Exponential_Responses

# Discharging




- the discharged current in opposite to that in charging, with an initially current of 50 mA.
- The p.d. across the capacitor decreases as it discharged.

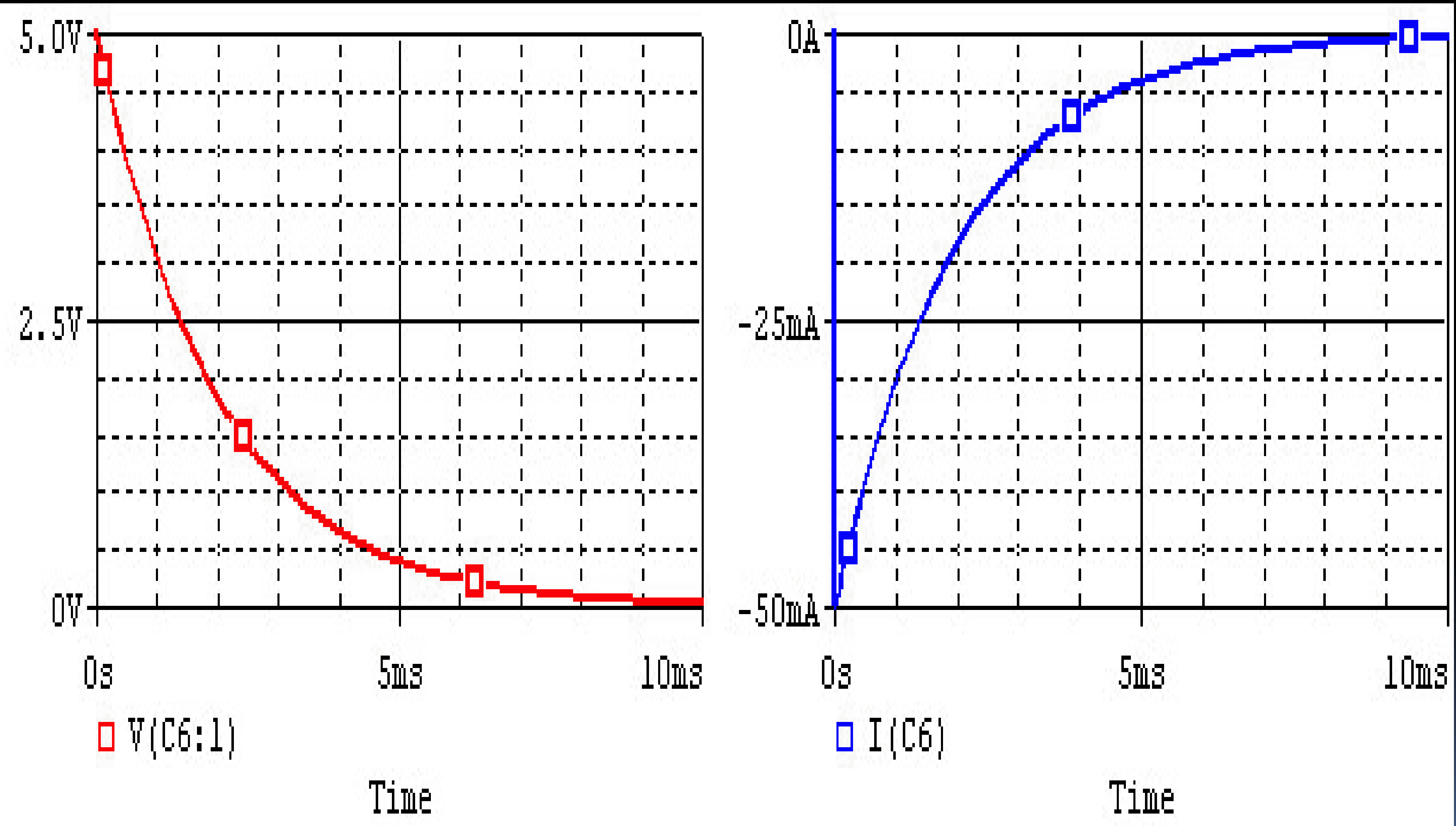
Consider the following circuit:

In the circuit, the capacitor is initially charged and has voltage $V_0 = 5$ V across it, and the switch is initially open. At time $t = 0$, we close the circuit and allow the capacitor to discharge through the resistor. The voltage across a capacitor discharging through a resistor as a function of time is given as:

$$v_C(t) = V_0 e^{-\frac{t}{RC}}$$

where $V_0$ is the initial voltage across the capacitor.

# Discharging



# Time constant, RC

- The term *RC* is the resistance of the resistor multiplied by the capacitance of the capacitor, and known as the *time constant*, which is a unit of time. The function completes 63% of the transition between the initial and final states at $t = 1RC$, and completes over 99.99% of the transition at $t = 5RC$.
- The voltage and current of the capacitor in the circuits above are shown in the graphs, from $t=0$ to $t=5RC$. Note the polaritiy—the voltage is the voltage measured at the "+" terminal of the capacitor relative to the ground (0V). A positive current flows into the capacitor from this terminal; a negative current flows out of this terminal.:

# 23.7 Charge and discharge of capacitors

In Figure 23.18, the capacitor is charged by the battery when the switch is connected to terminal P. When first connected to P, a current is observed in the microammeter. The current starts off quite large and gradually decreases to zero. When connected to terminal Q, the capacitor discharges through the resistor and a current in the opposite direction is observed. As with the previous current, it starts off large and gradually falls to zero.

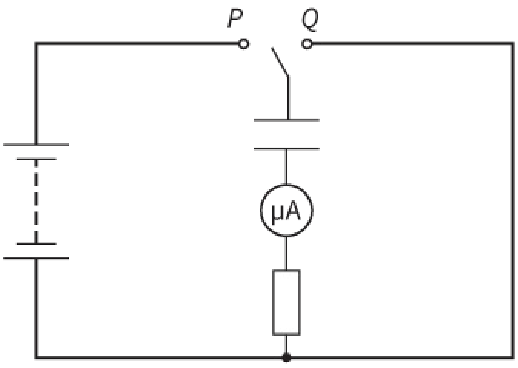



**Figure 23.18** A circuit to charge and discharge a capacitor.

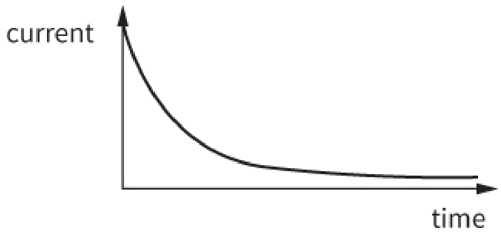



**Figure 23.19** A graph showing how the current changes with time when a capacitor discharges through a resistor.

This shape of this graph it is quite common in sciences and it occurs in different situations – you will come across it again in radioactive decay in Chapter 29. In this case, it comes from the fact that, as charge flows off the capacitor, the potential difference reduces and so the current (the charge flowing per unit time) in the circuit also decreases. In radioactive decay, it occurs because as atoms decay, there are fewer atoms left to Charles's law and, therefore, fewer decays per unit time.

This type of decay is called **exponential decay** and is described by the formula:

$$x = x_0 e^{-ky}$$

where $x$ is the dependent variable, $y$ is the independent variable, $k$ and $x_0$ are constants and $e$ is the exponential function (a naturally occurring number of value 2.7118 28 ...).

## Question

**22** In the circuit in Figure 23.18, the resistance has a resistance of 2000 Ω, the capacitor has a capacitance of 1000 μF and the battery has an e.m.f. of 12 V.

- **a** Calculate:
  - **i** the potential difference across the capacitor when it is fully charged by the battery
  - **ii** the charge stored by the capacitor when it is fully charged
  - **iii** the current in the resistor when the switch is first connected to terminal Q.
- **b** Explain what happens to the amount of charge stored on the plates in the moments after the switch is first connected to terminal Q.
- **c** Based on your answer to part b, explain what effect this has on:
  - **i** the potential difference across the capacitor
  - **ii** the current in the resistor.

Once you have worked through Question 22, you should understand why the current gradually reduces: it reduces because of the current itself, as it takes charge off the plates.

What is the effect of changing the resistance in the circuit? There will be no change in the initial potential difference across the capacitor, but the initial current through the resistor will be changed. Increased resistance will mean decreased current, so charge flows off the capacitor plates more slowly and, therefore, the capacitor will take longer to discharge. Conversely, decreasing the resistance will cause the capacitor to discharge more quickly.

What is the effect of increasing the capacitance of the capacitor? The initial p.d. across the capacitor is, again, unchanged. So, with an unchanged resistance, the initial current will be unchanged. However, there will be more charge on the capacitor and so it will take longer to discharge.

From this, we can see that the time taken for a capacitor to discharge depends on both the capacitance and the resistance in the circuit. The quantity $RC$ is called the **time constant** of the circuit. It is written using the Greek letter tau ($\tau$).

**KEY EQUATION**

$$\tau = RC$$

Time constant for a capacitor discharging.

## Question

**23** Show that the unit of the time constant ($RC$) is the second.

The equation for the exponential decay of charge on a capacitor is:

$$I = I_0 \exp\left(-\frac{t}{RC}\right)$$

where $I$ is the current, $I_0$ is the initial current, $t$ is time and $RC$ is the time constant.

The current at any time is directly proportional to the potential difference across the capacitor, which in turn is directly proportional to charge across the plate. The equation also describes the change in the potential difference and the charge on the capacitor.

So:

$$V = V_0 \exp\left(-\frac{t}{RC}\right)$$

where $V$ is the p.d, and $V_0$ is the initial p.d.

And:

$$Q = Q_0 \exp\left(-\frac{t}{RC}\right)$$

where $Q$ is the charge and $Q_0$ is the initial charge.

**KEY EQUATIONS**

**Exponential decay** of charge on a capacitor:

$$I = I_0 \exp\left(-\frac{t}{RC}\right)$$
$$Q = Q_0 \exp\left(-\frac{t}{RC}\right)$$
$$V = V_0 \exp\left(-\frac{t}{RC}\right)$$

**WORKED EXAMPLE**

**4** The potential difference across the plates of a capacitor of capacitance 500 μF is 240 V. The capacitor is connected across the terminals of a 600 Ω resistor.

Find the time taken for the current to fall to 0.10 A.

**Step 1** Calculate the initial current:

$$\begin{aligned} I_0 &= \frac{V}{R} \\ &= \frac{240}{600} \\ &= 0.40\ \text{A} \end{aligned}$$

**Step 2** Calculate the time constant:

$$\begin{aligned} \tau &= RC \\ &= 600 \times 500 \times 10^{-6} \\ &= 0.30\ \text{s} \end{aligned}$$

**Step 3** Substitute into the equation:

$$\begin{aligned} I &= I_0 \exp\left(-\frac{t}{RC}\right) \\ 0.10 &= 0.40 \exp\left(-\frac{t}{0.30}\right) \\ \frac{0.10}{0.40} &= \exp\left(-\frac{t}{0.30}\right) \\ 0.25 &= \exp\left(-\frac{t}{0.30}\right) \end{aligned}$$

**Step 4** $e$ comes from the antilog of the natural logarithm (ln) such that ln ($e^{x}$) = $x$

Taking ln of both sides:

$$\begin{aligned} \ln 0.25 &= -\frac{t}{0.30} \\ -1.386 &= -\frac{t}{0.30} \\ t &= 1.386 \times 0.30 \\ &= 0.41\ s \end{aligned}$$

## Question

**24** A 400 μF capacitor is charged using a 20 V battery. It is connected across the ends of a 600 Ω resistor with 20 V potential difference across its plates.

**a** Calculate the charge stored on the capacitor.

**b** Calculate the time constant for the discharging circuit.

**c** Calculate the time it takes the charge on the capacitor to fall to 2.0 mC.

**d** State the potential difference across the plates when the charge has fallen to 2.0 mC.

### REFLECTION

In Worked example 3, we showed that when a charged capacitor is connected to an identical uncharged capacitor, half the energy is dissipated in driving the charge through the circuit and is transformed to thermal energy. If we had superconducting connectors – ones that conduct electricity without any energy losses – what would happen? Discuss with a partner.

What did you find satisfying about discussing this problem?